ModelIt!

Teacher's Guide

Predicting the Next Pandemic

Predictive Modeling of Zoonotic Spillover and Pandemic Emergence

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS2-6, HS-ETS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-6, HS-ETS1-4

Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem; use a computer simulation to model the impact of proposed solutions to a complex real-world problem.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational simulation to model the probabilistic chain of events from zoonotic spillover to pandemic emergence, varying parameters to predict outbreak trajectories.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics / ETS1.B Developing Possible Solutions** — Changing conditions in ecosystems affect organism interactions; engineering solutions to pandemic preparedness require modeling complex, multi-variable systems.
Crosscutting Concept	**Cause and Effect / Scale, Proportion, and Quantity** — Students trace the causal chain from local wildlife contact to global pandemic, analyzing how small changes in early-stage parameters produce vastly different outcomes at global scale.

CAST Testing Alignment

MULTIPLE CHOICE:

A novel virus is detected in a remote village 4 weeks after the first case, compared to a scenario where it's detected in 3 days. Based on your model, how does this delay affect pandemic probability?

CONSTRUCTED RESPONSE:

Using your predictive model, explain why deforestation in tropical regions increases pandemic risk. Trace the full causal chain from land-use change to potential pandemic emergence, referencing at least three model components.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to predictive epidemiological modeling — the computational approach that can identify where and when the next pandemic is most likely to emerge. Biotech skill focus: Predictive modeling and optimization. COVID-19 demonstrated both the devastating consequences of failed pandemic prediction and the power of computational models to guide response. Students will build models that identify the conditions under which zoonotic spillover escalates to global catastrophe.

Zoonotic Spillover: Where Pandemics Begin

Approximately 75% of emerging infectious diseases are zoonotic — they originate in animals. HIV came from chimpanzees, Ebola from bats, SARS-CoV-1 from civets (originally bats), MERS from camels (originally bats), COVID-19 likely from bats (possibly through an intermediate host). The spillover process requires three conditions: contact (humans must encounter infected animals), compatibility (the virus must be able to infect human cells), and transmission (the virus must be able to spread between humans). Most zoonotic spillover events are dead ends — the virus infects one person and stops. But occasionally, mutation or recombination produces a variant that achieves sustained human-to-human transmission, and a pandemic is born. The PREDICT project identified over 1,000 novel viruses in wildlife, including 160+ coronaviruses, before it was defunded in 2019 — months before COVID-19 emerged.

The Pandemic Prediction Problem

Pandemic emergence is a low-probability, high-consequence event that follows a multi-step stochastic process. Each step has its own probability: the chance of a spillover event during any given contact (~0.001-1%), the chance that the virus achieves human-to-human transmission (~0.1-10% of spillover events), the chance of sustained transmission before containment (~10-50% depending on response speed), and the chance of global spread (~50-90% once sustained transmission is established in a connected population). Predictive models don't forecast specific pandemic dates — they identify the conditions, locations, and viral families where the compound probability is highest. This is analogous to earthquake prediction: we can't say exactly when, but we can identify the fault lines.

Surveillance: The Critical Intervention Window

The single most important determinant of pandemic prevention is surveillance speed — the time between the first human case and public health detection. COVID-19 circulated in Wuhan for an estimated 4-8 weeks before being identified and reported. By that time, the virus had infected thousands and spread to multiple countries through air travel. Models consistently show that detecting and responding to a novel pathogen within the first week of human cases — when the infected population is still small and geographically contained — can reduce pandemic probability by 90%+. Modern surveillance tools include wastewater genomic monitoring (detecting viral RNA in sewage before clinical cases are reported), syndromic surveillance (AI analysis of hospital admissions and symptom patterns), and metagenomic sequencing of clinical samples (identifying any pathogen, known or novel, directly from patient specimens).

Pandemic Risk Mapping: Predicting Hotspots

The geographic distribution of pandemic risk is not random — it clusters in predictable hotspots. The highest-risk zones are tropical and subtropical regions with high wildlife biodiversity (especially bat species diversity), rapid deforestation and agricultural expansion (increasing wildlife-human contact), dense human populations near wildlife interfaces, live animal markets, intensive livestock farming (particularly pigs, which can serve as 'mixing vessels' for influenza reassortment), limited public health surveillance infrastructure, and high international travel connectivity. South and Southeast Asia, Central and West Africa, and parts of South America are consistently identified as the highest-risk regions. Predictive models integrate these risk factors into geographic risk maps that can guide targeted surveillance deployment.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine components spanning ecological (Wildlife-Human Contact Rate), virological (Viral Mutation Rate, Zoonotic Spillover Probability), epidemiological (Population Density, Immune Naivety, Travel Connectivity), and public health (Surveillance Detection Speed, Healthcare Capacity, Intervention Speed) domains.

E **Establish Relationships**

Students determine that Wildlife-Human Contact Rate creates spillover opportunities, Viral Mutation Rate generates human-compatible variants, and the chain from Zoonotic Spillover Probability through Population Density and Travel Connectivity to global spread is modulated by Surveillance Detection Speed, Healthcare Capacity, and Intervention Speed.

V **Visualize & Model**

Students build a predictive model connecting the spillover-to-pandemic pathway, visualizing how changes in surveillance and connectivity parameters shift the probability of pandemic emergence.

E **Evaluate Outcomes**

Students run slow-burn, COVID-19-analog, and optimized-early-warning scenarios to identify the parameter combinations that produce pandemics versus contained outbreaks.

R **Revise & Extend**

Students add Seasonal Climate Patterns, Viral Recombination Events, or Global Supply Chain Disruption to explore more realistic and consequential pandemic dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"COVID-19 was not a surprise to epidemiologists — they had been warning about exactly this scenario for decades. The virus, the location, even the timing were all predictable in hindsight. Today you're going to build the model that predicts the next one."*

Watch For

Show a timeline of pandemic warnings that were ignored before COVID-19. Let students process the pattern.

Slide 2: Learning Objectives

3 min

Say: *"You're going to build a predictive model for pandemic emergence — the same type of computational tool that organizations like the WHO and CDC use to assess global health threats."*

Watch For

Pre-teach zoonotic spillover and immune naivety. Show a simple animation of a virus jumping from bat to human to illustrate the spillover concept.

Slide 3: Big Question

3 min

Say: *"Somewhere in the world right now, a virus is mutating inside an animal. One day — maybe tomorrow, maybe in ten years — it will jump into a human. What determines whether that becomes a footnote or a catastrophe?"*

Watch For

Think-pair-share: Students list conditions that would make a spillover event more likely to become a pandemic. Compile and categorize.

Slide 4: LEVER Framework

2 min

Say: *"We're building a predictive model — one that identifies WHERE pandemics are most likely to start and WHAT conditions enable their spread. This is proactive science."*

Watch For

Preview LEVER steps. Distinguish between prediction (knowing the future) and preparedness (being ready for it). Both require models.

Slide 5: Activity 1: Components

8 min

Say: *"Four of these components are things governments can actually control — investments they can make RIGHT NOW to prevent the next pandemic. Which four? And which five are determined by nature?"*

Watch For

Guide sorting of external versus internal components. Emphasize that the controllable variables represent policy decisions — funding surveillance, managing wildlife contact, regulating travel.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the pathway from a bat in a cave to a global pandemic. Every link in this chain is a potential intervention point — a place where we could break the chain if we were watching."*

Watch For

Students map the full causal chain. Identify which connections represent the best intervention points (surveillance, containment) versus immutable factors (viral mutation rate).

Slide 7: Activity 3: Simulation

10 min

Say: *"Run these three scenarios and find the answer: Can we prevent the next pandemic if we invest in the right interventions — or are pandemics inevitable?"*

Watch For

Students predict outcomes, then run all three scenarios. Compare pandemic probability across scenarios, focusing on how surveillance speed changes outcomes.

Slide 8: Discoveries

5 min

Say: *"Your model shows what epidemiologists have been saying for years: pandemics are predictable and preventable — but only if we invest in surveillance BEFORE the next one starts, not after."*

Watch For

Connect model findings to COVID-19 timeline. Show how early detection could have changed the trajectory. Discuss why surveillance funding was cut before the pandemic.

Slide 9: STEM Challenge

12 min

Say: *"The WHO just hired your team. Design the surveillance system that detects the next pandemic virus within 72 hours of the first human case. This is the most important engineering challenge of our generation."*

Watch For

Groups design early warning systems including data streams, detection algorithms, alert protocols, and deployment strategies for resource-limited settings. Present and evaluate.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Wildlife-Human Contact Rate, Surveillance Detection Speed, Population Density, Travel Connectivity
- **INTERNAL:** Viral Mutation Rate, Zoonotic Spillover Probability, Immune Naivety, Healthcare Capacity, Intervention Speed

Relationships

- **Wildlife-Human Contact Rate to Zoonotic Spillover Probability:** POSITIVE (+) — More frequent contact between humans and wildlife creates more opportunities for cross-species viral transmission — each contact event is a probabilistic opportunity for spillover.
- **Viral Mutation Rate to Zoonotic Spillover Probability:** POSITIVE (+) — Higher viral mutation rate generates more genetic variants, increasing the probability that a human-compatible variant emerges that can infect human cells and achieve human-to-human transmission.
- **Population Density to Intervention Speed:** NEGATIVE (-) — Higher population density makes containment more difficult because infected individuals have more contacts before isolation, increasing the number of cases that must be traced and requiring more resources for intervention.
- **Travel Connectivity to Geographic Spread Rate:** POSITIVE (+) — More transportation links from the outbreak origin accelerate the geographic spread of the pathogen to new population centers, reducing the time window for containment.
- **Surveillance Detection Speed to Intervention Speed:** POSITIVE (+) — Faster surveillance detection enables faster intervention — you cannot contain what you haven't detected. Every day of detection delay exponentially increases the infected population that must be contained.

Key Discoveries

- Zoonotic spillover events happen far more frequently than pandemics — most cross-species jumps lead to dead-end infections that don't transmit between humans, but each one is a roll of the evolutionary dice
- The critical window for containment is extremely narrow — once a novel virus achieves sustained human-to-human transmission in a connected population, pandemic spread is nearly inevitable with current infrastructure
- Surveillance detection speed is the single most impactful intervention point — the difference between detecting an outbreak in week 1 versus week 4 can mean the difference between 100 cases and 100,000 cases
- Pandemic risk is not random — it clusters in predictable geographic hotspots where high wildlife diversity, rapid deforestation, dense human populations, and limited healthcare infrastructure overlap

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Pandemics are unpredictable 'acts of God'"	Pandemics follow a predictable sequence: zoonotic spillover, adaptation to human transmission, spread through connected populations. The specific pathogen and exact timing are uncertain, but the geographic hotspots, risk factors, and transmission dynamics are well understood. Epidemiologists predicted a novel coronavirus pandemic from a bat reservoir in a densely populated Asian region years before COVID-19 — the surprise was not THAT it happened but that the world wasn't ready.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"COVID-19 was a once-in-a-century event"	The frequency of pandemic-potential zoonotic spillover events is INCREASING, not decreasing. Deforestation is accelerating wildlife-human contact. Climate change is shifting animal ranges. Global travel connectivity is increasing. Industrial livestock farming creates massive pathogen amplification opportunities. The conditions that produced COVID-19 are more common today than they were in 2019, making the next pandemic more likely, not less.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We just need better drugs and vaccines to stop pandemics"	Drugs and vaccines are response tools — they help AFTER a pandemic has started. By the time a vaccine is developed, tested, manufactured, and distributed (12-18 months minimum), millions may already be infected. The model shows that prevention through surveillance and early containment is orders of magnitude more effective and cheaper than response. The COVID-19 vaccine was the fastest ever developed, yet 5 million people died before it was widely available.